MIL-P-20304
21 November 1951

MILITARY SPECIFICATION

PENTAERYTHRITOL

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments, and supersedes the following specification:

Army 50-11-122
5 February 1944

This specification consists of this cover sheet and Specification 50-11-122, dated 5 February 1944, attached hereto, without modification.

When a request for this specification is received by a supplying activity it will be necessary to attach this cover sheet to the pertinent specification before issue.

Copies of specifications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer .

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GPO 83-22316

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U. S. ARMY
SPECIFICATION

No. 50-11-122
5 February 1944

PENTAERYTHRITOL

A. APPLICABLE SPECIFICATIONS

A-1. The following Specifications of the issue in effect on the date of invitation for bids, form a part of this specification.

Federal Specification No. RR-S-366	-	Sieves, Standard, Testing.
U. S. Army Specification No. 50-0-1	-	General Specifications for Ammunition, Except Small Arms Ammunition.
U. S. Army Specification No. 100-2	-	Standard Specifications for Marking Shipments.

B. GRADE.

B-1. This specification covers one grade of pentaerythritol.

C. MATERIAL AND WORKMANSHIP

C-1. There are no requirements with respect to material and workmanship applicable to this specification.

D. GENERAL REQUIREMENTS

D-1. See section E.

E. DETAIL REQUIREMENTS.

E-1. Color. White
E-2. Moisture. Maximum, 0.5 percent.
E-3. Melting point. Minimum, 250°C.
E-4. Hydroxyl value. Minimum 49.0 percent.
E-5. Water insoluble. Maximum, 0.25 percent
E-6. Chlorides. Maximum, 0.01 percent.
E-7. Ash. Maximum, 0.10 percent.
E-8. Granulation.

Through U. S. No. 18 sieve, minimum	100 percent
Through U. S. No. 20 sieve, minimum	99 percent
Retained on U. S. No. 100 sieve, minimum	90 percent

F. METHODS OF SAMPLING, INSPECTION AND TESTS.

F-1. Inspection. Inspection shall be as specified in Specification No. 50-0-1, General Specifications for Ammunition Except Small Arms Ammunition and shall be at the point of delivery unless otherwise specified in the contract.

F-2. Size of lots. Maximum, 30,000 pounds.

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F-3. Sampling shall be done as follows: Select 10 percent but in no case more than 10 of the containers comprising the lot, so as to be representative of the lot. By means of a scoop remove from each selected container sufficient of the material to form a primary sample of two pounds. Mix this sample thoroughly, place a one-pound portion in a rubber-stoppered bottle and label each primary sample so that the container from which it was taken can be easily identified. Mix the remaining portions of the primary samples thoroughly and quarter until a composite sample of about two pounds is obtained. Place the composite sample in a rubber-stoppered bottle and label so as to show the name of the material, manufacturer, plant, purchase order, and the number of pounds in the lot. All acceptance tests shall be made on the composite sample representative of the lot. Hold the primary samples for possible future examination should the composite sample fail to meet the requirements.

F-4. Laboratory examination shall be made as follows:

F-4a. Moisture. Transfer a weighed portion of approximately five grams of the sample to a tared glass weighing dish. Place the dish and sample in an oven at 110° to 120°C. for two hours, cool in a desiccator and weigh. Calculate the loss in weight to percentage of moisture in the sample.

F-4b. Melting point.

F-4b(1). Apparatus. Set up a melting point bath equipped with a mechanical stirrer and a source of heat that can be easily regulated. A beaker of one or two liters capacity three-fourths full of clean cottonseed oil is recommended. Suspend a Centigrade thermometer, accurately standardized for total immersion, in the bath so that the bulb is not less than 1.5 inches from the bottom of the bath. If the mercury column will not be completely immersed at the temperature of the observed melting point, suspend a second thermometer about 0.5 inch from the first thermometer with its bulb at about the height of the middle of the exposed mercury column of the first thermometer.

F-4b(2). Determination. Grind a small portion of the sample to a fine powder, and fill to a depth of about 10 mm. a thin-walled capillary tube of uniform diameter, long enough to extend above the top of the bath. Compact the material by tapping, and fasten the tube to the standardized thermometer so that the lower end of the tube is in contact with the bulb of the thermometer. Start the stirrer and heat the bath rapidly to 225°C. Then regulate the heat so that the temperature rise is between 1° and 2°C. per minute. Continue heating until the last crystals disappear in the clear liquid. Consider this temperature to be the melting point. This point is best determined with the aid of a magnifying glass and illumination from the rear.

F-4b(3). Correction. If the mercury column is completely immersed at the melting point, apply any necessary calibration correction. If part of the column is exposed, add the following correction:

$$\text{Correction} = N(T-t) \quad .000159$$

where

N = number of degrees in exposed mercury column
T = uncorrected temperature
t = average temperature of exposed mercury column

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F-4c. Hydroxyl value. Transfer an accurately weighed portion of approximately 0.5 gram of the dried sample to an Erlenmeyer flask to which a condenser may be fitted by means of a ground glass joint. By means of a pipette add a 25 ml. portion of pyridine-acetic acid solution which has been prepared as described in paragraph F-4c(1). Attach a condenser to the flask, place the flask on a hot plate, and reflux the solution gently for 30 minutes. Flush the condenser with 30 to 50 ml. of water, and cool the flask in an ice bath before removing and rinsing the glass joint. Titrate the solution at once with approximately normal sodium hydroxide solution, using phenolphthalein as the indicator. Make a blank determination on the pyridine-acetic acid solution by treating a 25 ml. portion in exactly the same way. Calculate the hydroxyl value as follows:

$$\text{Percent hydroxyl} = \frac{1.7 \times N \times (A-b)}{W}$$

where

N = normality of sodium hydroxide solution.
A = number of ml. of sodium hydroxide solution used in titrating sample
B = number of ml. of sodium hydroxide solution used in titrating blank
W = weight of sample.

F-4c(1). Prepare the pyridine-acetic acid solution as follows: Add exactly 3.5 ml. of water to 1000 ml. of pyridine which has been previously dried over solid potassium hydroxide for at least 72 hours. Mix the solution and add 140 ml. of acetic anhydride. It is important that both the amount of water and the order of addition specified be strictly adhered to, otherwise low results are liable to be obtained. The presence of 0.35 percent of water inhibits the formation of dark colored resins in the blank during the refluxing period due to the reaction between dry pyridine and acetic anhydride. However, since the addition of more than 0.4 percent of water causes low results on the acetylat on, the pyridine is first dried, and then the exact amount of water specified is added. The water must be added to the pyridine before the addition of the acetic anhydride; otherwise low results will be obtained.

F-4d. Water insoluble. Transfer a weighed portion of approximately 15 grams of the sample to a one-liter Erlenmeyer flask. Add 500 ml. of distilled water, measured with a graduated cylinder. Close the flask with a rubber stopper, and shake the flask and contents at 15-minute intervals over a period of one hour. Allow the flask and contents to stand for at least 16 hours at room temperature. Filter the solution through a tared Gooch crucible, wash any residue with several portions of water at 25°C. Dry the crucible for one hour at 100°C. cool in a desiccator, and weigh. Calculate the increase in weight to percentage of insoluble matter in the sample.

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F-4e. Chlorides. Transfer a weighed portion of approximately 10 grams of the sample to a beaker and dissolve in 300 ml. of distilled water. Add 5 ml. of approximately 40 percent formaldehyde and 5 ml. of concentrated nitric acid. Cover the beaker with a watch glass, and boil the solution for 10 minutes. Remove the beaker from the source of heat, and add approximately 1 ml. of a 10 percent silver nitrate solution dropwise and with constant stirring. Heat the mixture to the boiling point and allow to boil until the precipitate coagulates. Filter the solution through a tared Gooch crucible, wash the precipitate with 1 percent nitric acid solution until the washings give no precipitate when tested with hydrochloric acid, and then wash twice with distilled water. Dry the crucible and contents for 10 minutes at 100°C. and for 1 hour at 130° to 150°C., cool in a desiccator and weigh. Calculate the increase in weight to percentage of chlorides in the sample.

$$\text{Percentage of chlorides, as chlorine} = \frac{24.74\ A}{B}$$

where

A = weight of silver chloride
B = weight of sample

F-4f. Ash. Transfer a weighed portion of approximately 10 grams of the sample to an ignited and tared platinum or porcelain dish. Ignite the material over a low Bunsen Flame and allow to burn slowly without further heating. After the flame dies down, place the dish in a muffle furnace at 800°C. for approximately 30 minutes, until all carbonaceous material is consumed. Cool the dish in a desiccator and weigh. Calculate the increase in weight to percentage of ash in the sample.

F-4g. Granulation. Fit a No. 18 U. S. Standard sieve on a No. 20 U. S. Standard sieve and this on a No. 100 U. S. Standard sieve provided with a bottom pan. Place a weighed portion of approximately 50 grams of the sample on the No. 18 sieve, cover the assembly, and shake for 15 minutes. Weigh the material retained on each sieve. Calculate the percentage passing through the No. 18 and No. 20 sieves, and the percentage retained on the No. 100 sieve.

G. PACKAGING, PACKING AND MARKING

G-1. Packaging. Unless otherwise specified commercial packages are acceptable under this specification.

G-2. Packing. Unless otherwise specified, pentaerythritol shall be delivered in standard commercial containers so constructed as to insure acceptance by common or other carriers, for safe transportation, at the lowest rate to the point of delivery.

G-3. Marking. In accordance with U. S. Army Specification No. 100-2.

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H. NOTES

H-1. Uses. Pentaerythritol covered by this specification is intended for use in the manufacture of pentaerythrite tetranitrate (PETN).

H-2. The use of this specification, whenever applicable, is mandatory on all procuring agencies of the Army.

H-3. If the composite sample representative of the lot fails to pass the inspection tests, the manufacturer shall have the option of having analysis of each primary sample made at his own expense. The manufacturer may then replace defective portions of the lot represented by the primary samples which fail to meet the requirements, and again submit the lot for inspection and acceptance, provided that the markings on the container are such that complete replacement of defective portions of the lot can be made to the satisfaction of the inspector.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact the Government may have formulated, furnished, or in any way supplied the said drawing, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation or conveying any rights or permission to manufacture, use, or sell any patented invention that may be in any way related thereto.

Note: Copies of this specification may be obtained from the Office, Chief of Ordnance, Army Service Forces, Washington, D. C.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

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